

12-Week Internship Schedule

Python Programming & AI/ML

This 12-week internship program is designed for Diploma Engineering students to develop practical skills in Python Programming, Data Science, and Machine Learning. The internship focuses on hands-on learning, real-world data problem-solving, building predictive models, and industry-oriented application deployment.

Internship Details

Particular	Details
Internship Title	Python Programming & AI/ML Internship
Duration	12 Weeks
Target Audience	Diploma Engineering Students
Mode	Hybrid (Online + Offline Practical)
Sessions Per Week	3-4 Sessions
Session Duration	1.5-2 Hours
Learning Approach	Theory + Hands-on + Projects
Final Outcome	Industry-Level AI/ML Capstone Project

Internship Objectives

- Understand Python programming fundamentals and advanced best practices.
- Develop logic-building, problem-solving, and error-handling skills.
- Perform comprehensive Exploratory Data Analysis (EDA) and data manipulation.
- Build, train, and evaluate predictive models using regression and classification.
- Develop an optimized, end-to-end machine learning pipeline.
- Deploy functioning local web applications featuring trained ML models.
- Create an industry-level mini-project and present a live capstone app.

Tools & Technologies Covered

- **Programming:** Python
- **Data Manipulation:** NumPy, Pandas
- **Data Visualization:** Matplotlib, Seaborn
- **Machine Learning:** Scikit-Learn, TensorFlow/Keras, PyTorch
- **Deployment & UI:** Streamlit, Gradio
- **Version Control:** GitHub

Internship Structure

Students will participate in a structured environment treating them like real software engineers. The program culminates in an end-to-end Capstone Project that must include a real-world dataset, data visualization, cleaned data with trained models, and a final Streamlit web application deployment hosted on GitHub.

Week-Wise Internship Schedule

Week 1: Python Fundamentals Variables, data types, and operators. Control flow (if/else, loops) and functions. Data structures (lists, dictionaries, sets, tuples). **Outcome:** Students will deliver a basic logic-based Python script (e.g., a text-based calculator or password generator).

Week 2: Advanced Python & Best Practices Object-Oriented Programming (OOP) basics including classes and objects. Error/Exception handling (try-except) and File I/O (reading and writing CSV/JSON files). **Outcome:** Students will deliver a script that reads a local file, processes the text/data, and writes the output to a new file.

Week 3: Data Manipulation Engine Introduction to NumPy (arrays and mathematical operations) and Pandas (Series, DataFrames, filtering, and grouping data). Handling missing values and basic data cleaning. **Outcome:** Students will clean a raw, messy dataset from Kaggle.

Week 4: Exploratory Data Analysis (EDA) & Visualization Data visualization using Matplotlib and Seaborn. Understanding correlations, distributions, and outliers. **Outcome:** Students will perform a complete EDA on a dataset (like Titanic or Housing) and present 3 key business insights with charts.

Week 5: Intro to ML & Regression Supervised vs. Unsupervised learning. Simple and Multiple Linear Regression. Train/Test splits and understanding metrics (MSE, R-squared). **Outcome:** Students will deliver a model that predicts a continuous number (e.g., predicting house prices).

Week 6: Classification Models Logistic Regression, Decision Trees, and Random Forests. Evaluating classification models: Confusion Matrix, Accuracy, Precision, Recall. **Outcome:** Students will deliver a model that categorizes data (e.g., predicting customer churn or diabetes onset).

Week 7: Unsupervised Learning Clustering data using K-Means. Dimensionality reduction (PCA - taught conceptually). **Outcome:** Students will group a customer dataset into distinct marketing segments.

Week 8: Model Optimization & Pipeline Building Hyperparameter tuning (GridSearchCV). Feature engineering and scaling (StandardScaler, MinMaxScaler). **Outcome:** Students will build an optimized, end-to-end machine learning pipeline that beats a baseline score on a given dataset.

Week 9: Introduction to Deep Learning & Modern AI Conceptual overview of Neural Networks. Building a basic multi-layer perceptron using TensorFlow/Keras or PyTorch. Brief overview of how Large Language Models (LLMs) and prompt engineering work. **Outcome:** Students will deliver a simple neural network classifying image data (like the MNIST dataset).

Week 10: Model Deployment & UI Introduction to Streamlit or Gradio. Saving and loading trained models (Pickle or Joblib). Connecting the Python ML model to a web interface. **Outcome:** Students will deliver a functioning local web app where a user inputs data and gets a prediction.

Week 11: Capstone Project Development Students form teams or work individually on a unique problem. Ideation, data gathering, EDA, and model training. **Outcome:** Students will build their final project.

Week 12: Deployment, Documentation, and Presentation Deploying the Streamlit app to Streamlit Community Cloud (free and easy). Writing a clean README.md for their GitHub repository. **Outcome:** Final Deliverable: Demo Day! Students present their live web app and explain their methodology.